Chapter 29 worked solutions

Scientific Sequences and Biological Structure

Exercises 1–4 work through aligned sequences, residue features, and contact maps by hand. Exercises 9–11 design evaluations for protein representations and structure predictions.

1. Read aligned columns

In the alignment, read columns 2 and 4 from the same row. The first row is `ACDG`, so its pair is `(C, G)`.

Alignment row	Sequence	Pair from columns 2 and 4
1	ACDG	(C, G)
2	ACEG	(C, G)
3	ASDA	(S, A)
4	ASEA	(S, A)

The observed counts are **2 for `(C, G)` and 2 for `(S, A)`**. Both `(C, A)` and `(S, G)` have count zero.

Imagine rows 1 and 2 descend from a common ancestor that already had C and G at those positions, while rows 3 and 4 inherited S and A from another branch. The alignment would show the same association even if the two residues were far apart in the folded proteins. Each branch could pass down both letters regardless of whether the residues interact.

2. Preserve some order

Both `CAAC` and `CCAA` contain two A residues and two C residues. Composition divides each count by the sequence length. For each sequence the A and C entries are both $2/4 = 1/2$, and the other entries are zero.

For the adjacent pairs, slide a window of two letters one position at a time and read each pair in chain order. Each sequence of length four yields three pairs, so divide each pair count by three.

Sequence	Adjacent pairs in order	Nonzero normalized counts
CAAC	CA, AA, AC	CA $= 1/3$, AA $= 1/3$, AC $= 1/3$
CCAA	CC, CA, AA	CC $= 1/3$, CA $= 1/3$, AA $= 1/3$

In the pair matrix the row is the first letter and the column is the second, so `CA` and `AC` occupy different entries. The two sequences share one `CA` and one `AA`, but `CAAC` has an `AC` where `CCAA` has a `CC` in its counts. The pair frequencies distinguish the sequences.

3. Pool only residues

The residue mask is `[False, True, True, False]`, which selects $(2, 0)$ and $(0, 4)$. Add the vectors coordinate by coordinate and divide by the two selected rows.

$$\frac{(2, 0) + (0, 4)}{2} = (1, 2).$$

Including the beginning and ending markers gives a different average.

$$\frac{(10, 10) + (2, 0) + (0, 4) + (-10, -10)}{4} = \frac{(2, 4)}{4} = (0.5, 1).$$

The marker vectors cancel in the sum, but they still increase the denominator from two to four. The mask makes both the sum and its denominator depend only on actual residue rows.

4. Read a geometric threshold

Number the residues 1, 2, and 3 in the order given. A contact must pass two tests. Its Euclidean distance, the straight-line distance between the points, must be at most 3. Its sequence separation, the absolute difference of its indices, must be at least 2.

Unordered pair	Index separation	Distance	Accepted?
$(1, 2)$	1	2	No. The residues are adjacent.
$(1, 3)$	2	3	Yes. Both cutoffs include equality.
$(2, 3)$	1	$\sqrt{2^2 + 3^2} = \sqrt{13} > 3$	No. The pair fails both tests.

The only accepted unordered pair is $(1, 3)$. It sets both matrix entries $(1, 3)$ and $(3, 1)$ to 1 because distance and index separation are the same in either direction. All other entries are zero in the Boolean contact map. They would appear as dashes in the chapter's figure convention because they fail the index-separation rule, including the diagonal, where the separation is zero.

9. Split by relatedness

Suppose 40 proteins labeled *nuclear* belong to one family. They may share a motif or much of their sequence. A random split by protein can put close relatives on both sides, so a classifier may learn patterns specific to that family. Testing on other members of the family gives limited evidence about transfer to a new one.

Use each family as a group. Assign complete families to training, validation, or test, with every sequence from a family in just one split. For example, aim for 70 percent of families for training, 15 percent for validation, and 15 percent for test, subject to having enough families for the location labels being evaluated. A family can contain several locations, so check the actual family and sequence counts per label after assigning groups. Fit preprocessing and classifier parameters on training families, choose settings on validation, and evaluate once on test families. Predefine which labels contribute to the validation and test macro averages. Each scored label should be represented in training and in the respective held-out split.

Record the family assignment source and its version. Screen for identical sequences and close relatives that happen to have different family labels. If two nominal groups share close relatives under the study's identity and alignment-coverage rule, keep the linked groups together or omit the problematic records under a rule fixed before scoring. Report that rule and inspect the closest cross-split pairs. A family split tests transfer under a specified notion of relatedness.

If a location is represented by only one family, that family cannot supply labeled examples both for training and for a family-disjoint test. One defensible choice is to keep it entirely in training, where the classifier can learn that label. Then mark its test recall and F1 as unavailable for the family-transfer question and define the test macro average over the classes with held-out reference support. Predictions of the unsupported class on examples from other labels still count as errors for those examples. Publish the per-class support and the list of classes included in the macro score. The test estimates performance for locations represented by held-out families. More families with the single-family label would be needed to measure transfer for it.

10. Test what an embedding adds

Choose a labeled location dataset and keep the same family-disjoint training, validation, and test sets for all three representations. Resolve duplicate sequences and decide how to handle ambiguous residues and incomplete chains before the comparison. If a long sequence would exceed the frozen encoder's input limit, define a common eligible population in advance and report the excluded sequences. Each comparison should score the same test proteins.

For the **composition** baseline, divide the count of each amino acid by sequence length. For **dipeptides**, count each ordered adjacent pair and divide by the number of adjacent pairs. For a single residue, use the all-zero pair vector from the chapter. For the **frozen encoder**, choose a specific checkpoint and layer. Compute one vector per residue, then average the real residue rows. Use a mask to exclude padding and beginning and ending markers. A residue's vector can depend on the sequence around it even after the vectors have been averaged.

Fit the same classifier family, such as regularized multinomial logistic regression, on each representation. Learn feature scaling and classifier parameters using training rows only. Choose regularization on the same validation set and give each representation the same search budget. Record feature widths and report per-class scores and macro-F1 on the same held-out test sequences. The result compares complete feature representations, including differences in width and in what each representation learned before this labeled task.

For the pretrained encoder, record the checkpoint, its training corpus and version if documented, and any available time cutoff. When the corpus is accessible, compare sequence identifiers and exact sequence hashes and search for close relatives with a documented alignment method and identity and coverage rule. Classify each test sequence by its strongest evidence of exposure. Report exact matches, nonexact close matches, no match detected in the searchable corpus, and unchecked cases. With no access to the pretraining sequences, state that the overlap is unknown. A split of the labeled task alone cannot establish which sequences the encoder saw during pretraining.

Check **sequence length** explicitly. Train a length-only classifier from the same family using log length as its input. Define length bins on training data, then compare all three representations on the same test sequences within each bin. Report per-bin sample and class counts and use the same supported classes for all three representations in each bin. You can also append the same length feature to each representation and repeat the comparison. If the embedding advantage appears within similar-length groups and persists when the other baselines get length, that supports an explanation beyond length alone. A gain limited to one length range calls for a narrower claim.

11. Assess a structural claim

Imagine that one domain contributes a pocket wall and a second domain supplies a residue that contacts the ligand. A change in the angle between the domains can move that residue away while leaving both pocket walls locally well folded. The question therefore concerns relative placement as well as local structure.

Inspect **pLDDT**, AlphaFold2's per-residue estimate of local structural confidence, at the proposed contact residues in each domain and at the linker that joins them. High local scores can support the shape of each domain. Then inspect **predicted aligned error** (PAE), which estimates positional error for one residue when aligning on another. Check the blocks within each domain and both directions of the cross-domain block, especially for pairs that define the proposed site. Low within-domain error alongside high cross-domain error means the model is more confident about the separate domains than about their relative placement. The predicted pocket is a hypothesis to test. AlphaFold guidance on local confidence, AlphaFold guidance on aligned error

Separate two experimental questions. To ask whether the ligand binds the full-length protein, use a direct binding measurement such as isothermal titration calorimetry, which measures heat associated with the interaction. Include a background control and enough replication to distinguish binding from noise. A binding measurement supports an interaction under those conditions. To locate the interaction, seek an experimental structure of the ligand-bound protein with both domains visible. For example, a crystal structure with ligand density and adequate local resolution can show where the ligand sits and which domain residues lie near it. Inspect whether the experimental density supports the ligand and nearby side chains.

Test candidate contacts by changing selected residues on each side of the proposed site and measuring binding again. Include the unmodified protein and a control change away from the site. Check that the modified proteins remain folded and in a comparable assembly. A change in binding could come from a disrupted fold or domain orientation as well as from loss of a direct contact. A ligand-bound structure that places the ligand between the two domains and controlled binding changes on both sides provide complementary evidence. A study of SlyD illustrates how structural and binding experiments can be combined for a two-domain protein. Quistgaard et al., 2016